EL TORBELLINO

NEWSLETTER OF SAN DIEGO ORBITEERS FREE FLIGHT CLUB

June 2026

The Prez's Corner *– Don Bartick*

Now that we're flying both Indoors and Outdoors makes you think that our Free Flight hobby is still alive and well. Well think again. The Perris field is iffy. No one seems to know how much longer the field will be available. The terrain is difficult to chase planes on foot. For me and other older people, only one event is doable. If we had flat trails going in various directions that would be friendly to pedal bikes, electric bikes and gas trail bikes, the chase would be significantly improved. At least we could get close to our downed planes with a lot less effort. Something for our field lessor to chew on. Furthermore, it appears the crop grown on the field is only good for the goats. We see them clearing the field. This is another hazard for us. So, if outdoor free flight is to survive, we need to find a field that can be groomed to be chase friendly. The Orbiteers have tried, but around San Diego, everything is being developed. But the I-15 corridor seems to have endless fields. If anyone reading this page have any ideas or knowledge to explore these open fields for availability, please take it upon yourself to do so. It's for the good of the hobby. Should you come across a candidate field, please advise me or one of our board members. The El Torbellino has a list of us in each publication.

On a better note, Arline and I have made plans to go back to Muncie, ID to participate in the 100th NATS, July 20 -24. It also gives us an opportunity to visit the National Model Airplane Museum to see, in person, the P-30 Exhibit I spearheaded.

In closing, I want to once again appeal to our members to seriously consider Indoor Free Flight. We have a good venue. Plus, Indoors has perfect weather, short chases and great competition. Shortfall: we need more participants to help defer the costs.

That's a wrap.

Remember: "Discipline is the bridge between goals and accomplishments"
Jim Rohn

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Non-Member Newsletter Subscription - $15
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Submit Dues to Club Treasurer:
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THE FINE PRINT **THE FINE PRINT**

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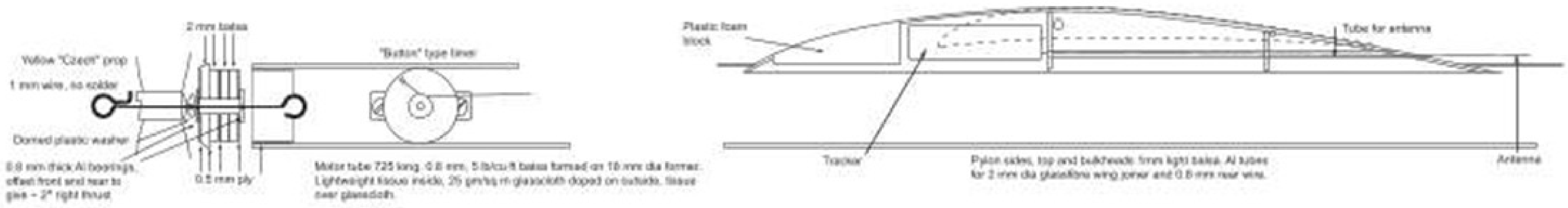
A Popular P-30 Design from Down Under
By Mike Jester

A free flight enthusiast from Australia recently sent me the plan for the A P-30 by Vin Morgan which is reproduced at the end of this article. He said it is a popular P-30 design flown by experienced fliers down under.

The A P-30 model is similar to the Polecat MK X P-30 designed and flown by Don DeLoach. He flies it with a 6 x 3/32-inch 10-gram rubber motor that provides a very long motor run. The A P-30 takes this to the extreme by using a 4 x 1/8-inch 10-gram rubber motor. Flying the A P-30 with a relatively thin rubber motor will only be successful in calm conditions and only if the model weighs very close to the 40-gram minimum. Too often a 4 x 1/8-inch rubber motor may not generate enough launch torque to break through ground turbulence and the model will land well under a 120 second max with many unused turns. Most P-30 fliers use a 6 x 1/8-inch 10-gram rubber motor. It produces a rocket climb but a relatively short motor run. The 1- mm (0.039-inch) prop shaft used in the A P-30 is way too delicate and will likely get bent on the first hard landing. It’s better to use a 0.047-inch prop shaft, and even better to use a 0.055-inch prop shaft in a P-30 model. The performance of the A P-30 could be improved by using a 9 1/2-inch Gizmo Geezer (GG) prop assembly. Its spring tensioner would allow the "hook-to-peg" distance to be greatly reduced without risking motor bunching and CG shifting. This will allow the wing to be positioned much further forward yet still achieve the optimum 66% CG indicated on the plan. Moreover, the increased separation between the wing and stab will allow the size of the fin to be reduced, resulting in less weight and less drag. The huge built-up fin on the A P-30 is removable and held onto the fuselage with a rubber band. Too many things can go wrong with a removable fin. A smaller fin made of 1/16-inch sheet balsa wood can be glued to the fuselage.

The A P-30 design has an unusual trim with the "*wing and thrustline are set parallel to the fuselage giving a few degrees of upthrust*." This trim will be advantageous in producing a better cruise with a weak rubber motor. However, it will likely present problems with a thicker rubber motor as needed to break through ground turbulence. Some down thrust will be needed if the A P-30 model is flown with a thicker rubber motor. The right/left trim pattern of the A P-30 is unconventional for a sport model with a fixed (non-folding) prop. A P-30 model should preferably climb right and glide right, otherwise there will be increased drag from the free-wheeling prop. Modern Coup and Wakefield models can use a right/left trim pattern because they have folding props. I would use a BMK RDT instead of a viscous timer. Then you can leave off the RF locator and its battery since the flier can DT the model after it reaches a max on one of the three official flights or on a fly-off. I think I might be able beat Vin Morgan in a P-30 contest with one of my simple Three Nite P-30 models, especially if his A P-30 model was launched in more than 3-4 mph of wind.

Plan for the A P-30 follows on next page.



A P-30

By Vin Morgan

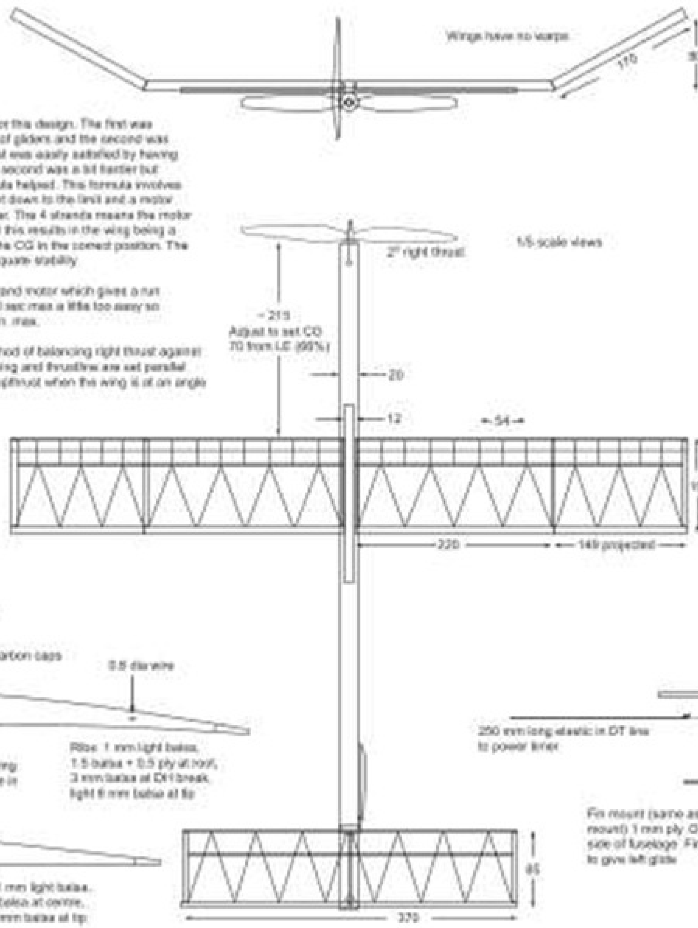
There were two principal requirements for this design. The first was that it should fit into a box with a couple of gliders and the second was that it should fly reasonably well. The first was easily satisfied by having two-piece wings and removable fin. The second was a bit harder but following the NSWFFS/Terry Bond formula helped. This formula involves a model with moderate wing area, weight down to the limit and a motor consisting of 4 strands of 1/8 tan 2 rubber. The 4 strands means the motor must run the full length of the model and this results in the wing being a relatively long way back in order to get the CG in the correct position. The large tail however seems to provide adequate stability.

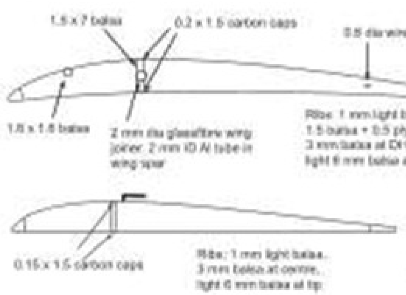
It is possible to get 2000 turns on a 4-strand motor which gives a run of nearly 50 seconds. This makes the 30 sec max a little too easy so the NSWFFS and the AFFS fly to a 2 min. max.

Trim is right/left by the conventional method of balancing right thrust against left rudder. There is no wing twist. The wing and thrustline are set parallel to the fuselage giving a few degrees of upthrust when the wing is at an angle of attack in flight. This helps to hold the nose up at the start of the climb while avoiding stalling tendencies at the end of the run when the power is reduced.

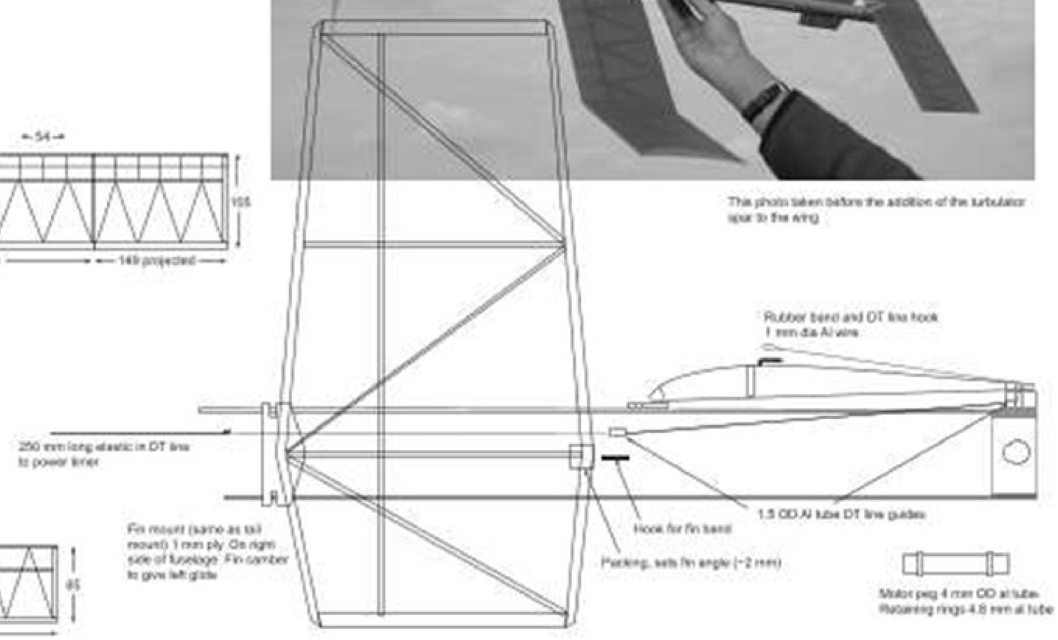
Weights

Prop and noseblock	9.3
Fuselage, tracker, timer	16.9
Fin	0.7
Tail	3.4
Wings	10.4
Wing joiner	0.2
Total	40.9 gm





This photo taken before the addition of the turbulator spar to the wing



Orbiteers May Results

COUPE-
1st Greg
2nd Dave
3rd Mike P

HLG-
1st Stan
2nd Chris
3rd John M

TOWLINE-
1st Clint
2nd Stan

Power-
1st Mike P
2nd Don
3rd Clint

Photo’s by Arline Bartick

Return to Indoor Flying---Don Bartick

June 7th, 2026 Indoors at the Los Coches Creek Middle School gym. The 2 competitions were Limited Penny Plane and Phantom Flash

Seven flyers registered and all 7 entered the events. 4 in Limited Penny Plane and 5 in Phantom Flash. The best 3 of 6 official attempts were scored and added together. The results were:

Limited Penny Plane

1st John Alling	851 seconds
2nd Mike Jester	612 seconds
3rd Don Bartick	194 seconds
3rd Robert Huffstutler	169 seconds

Phantom Flash

1st Walter Ainslie	272 seconds
2nd Greg Hutchison	271 seconds
3rd William Scott	209 seconds
4th Mike Jester	186 seconds
5th Don Bartick	158 seconds

The Penny Planes were affected by the A/C units running. They were supposed to be turned off. I followed up on this issue with the Cajon Valley Union School District the day after the event. As it turned out the Maintenance Supervisor forgot to issue a work order to turn the A/C units off. She was very apologetic and said she would not charge us for the gym rental the next time we schedule. We would still have to pay for custodial services. This would bring the rental fee down to $169.75 in place of $318.55.

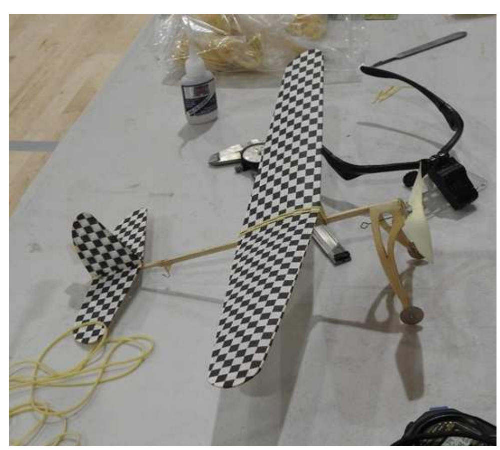
For this event, the 7 participants contributed $300. I ask for a minimum contribution of $30 for the 4-hour session. As you can see, the participant reached deep in their pockets to have the opportunity to fly indoors. But we need more participation. Gas alone to go to Perris to fly outdoors costs $50 in fuel If you're willing to travel to Perris to fly outdoor FF knowing the weather or field conditions could be iffy, then you should be willing to go to El Cajon to fly indoors knowing that there isn't an issue with weather of field conditions. Plus, the chases are darn easy. The challenge, in my opinion is more difficult.

Next Indoor is scheduled for August 2nd for 4 hours-8am to noon. Minimum contribution to participate is $30. Additional contributions are welcomed or may be needed. Competition events are free. Events will be A-6 and No-Cal. Of course, bring whatever you have to fun fly.

Up, up and away!
Don

Photo's by
Arline Bartick →

June 2026 Return to Indoor Flying Pictures - Arline Bartick



Goats, Dust Devils, and the May Scale Staffel

By David Lofthouse & William Scott

Attendance at this year's May Scale Staffel at Perris was light, but what it lacked in numbers it made up for in fellowship — and livestock.

The weather cooperated nicely both days. Cool mornings with a cloud layer that burned off around 10:30, reasonable winds, and just enough occasional dust twisters to keep things interesting.

The contest featured eleven events in all. The mass launches are always the crowd favorite — WWI, WWII, Greve/Thompson, and on Sunday, just as the thermals were starting to build, a Phantom Flash mass launch. Patrick's green Phantom Flash climbed out beautifully, then turned tail and headed downwind with purpose. Patrick and Walter gave chase, needing to hustle just to keep it in sight. About 45 minutes later they returned — model intact.

WWI Mass Launch — pilots with their aircraft at the Perris field

WWII Mass Launch — five pilots ready for battle

Upon arrival, we discovered the field had hired help. A roving herd of goats was hard at work managing the foxtail situation at Perris. If you've flown there, you know the foxtail. It's so thick and so uniform you'd swear someone planted it. I'm told that's unlikely. The goats don't seem to care either way — they were working it with enthusiasm. My ankles, by end of day, confirmed there's plenty left for next time.

The field's four-legged foxtail management crew — hard at work

I entered the Embryo event with my trusty Debut — a model that has seen a rebuild or two but still flies with dignity. My first official flight of the day was a thing of beauty. A full three minutes of graceful downwind drift, the DT having failed, settling the model directly into the densest concentration of goats on the property. Of course it did.

Turns out there was a mobile water tank nearby with a trough, which explained why that particular patch of real estate was prime goat territory. I trudged over to assess the situation. Two Great Pyrenees emerged from the herd with a professional level of concern about my intentions. I reconsidered the fence line. They slowed. I waited. Eventually I ventured back in, reminding myself that guard dogs aside, an inattentive modeler can also collect a solid head butt from an unwatched goat.

The goats, to their credit, were entirely indifferent to me. They parted just enough to let me search. I found nothing — no airframe, no gizmo prop, no viscous timer. Eaten or thoroughly trampled.

It brought to mind a story from my late friend John Hutchison, who once found himself searching for a model in a field of cows. He worked the herd patiently until he spotted one of them chewing contentedly — a braided motor dangling from its lips. The hobby has always had its hazards. Apparently some of them have four legs and a good appetite.

The Scale Staffel was a fine day regardless. Good people, good air, and a field crew that works for foxtails.

See you at the next one.

Mike Mulligan — 2026 Grand Champion →

San Diego Orbiteers
Flying Schedule 2026
Taibi Field Perris, Ca

Primary Date	Rain Date	Event
July 19	July 26	P-30/Glider/Power
August 16	Aug 23	Coupe/Glider/Power
September	(TBD)	**Free Flight Championships, Lost Hills, Ca**
September 20	September 27	OT/NOS Rubber/Glider/Power
October 18	October 25	P-30/Glider/Power
November	13,14&15	**Dual Clubs, Lost Hill, Ca**
November 22	November 29	Coupe/Glider/Power
December 13	No Date	Make-up

MP
12/25

COMMENTARY

UP FRONT

DYLAN VOLANTH

Fertile Ground

Aviation makes underappreciated **contributions to global agriculture**

AIR TRACTOR/GETTY IMAGES

WHEN AVIATION ANALYSTS OBSESS over the latest narrowbody backlog or business jet delivery forecast, they are overlooking a fleet of nearly 4,000 turbine-powered agricultural aircraft in operation around the world. This fleet protects crops worth hundreds of billions of dollars annually, yet it does not appear in major industry forecasts.

More than half of these aircraft fly in the U.S. in a sector that generates an estimated $37 billion in annual economic value across just five major crops—corn, soybeans, wheat, cotton and rice—according to the National Agricultural Aviation Association (NAAA). The NAAA reports that these aircraft treat 127 million acres of U.S. cropland annually, representing 28% of all cultivated U.S. acreage. Without aerial application of pesticides, American farmers would lose 1.7 billion bushels of corn and 295 million bushels of soybeans and require an additional 27 million acres of farmland, an area the size of Tennessee.

The fleet is dominated by Air Tractor and Thrush Aircraft, U.S.-based manufacturers that produce new, purpose-built agricultural aircraft priced between $1 million and $2 million. Air Tractor has been delivering more than 150 aircraft annually since 2021, underscoring the sector's robust demand. Its workhorse AT-502 represents 38% of the turbine fleet, followed by the larger AT-802 at 28%, both powered by variants of Pratt & Whitney Canada's PT6A. These are not vintage crop dusters; they are sophisticated platforms equipped with GPS guidance, variable-rate flow controllers and real-time meteorological systems that together achieve application accuracy that would be impossible with ground equipment.

The level of skill required to fly these aircraft rivals that in other aviation sectors. Agricultural aircraft pilots average nearly two decades of experience, and they execute 30-100 takeoffs and landings daily from rough airstrips, fly at 50 ft. while navigating obstacles and managing precise chemical application rates. The operational complexity explains why only a limited set of dedicated agricultural pilots serve the entire U.S. market. It takes significant investment to recruit candidates with both bush pilot skills and chemical application experience.

Why are aircraft used to protect agriculture? Three factors make them irreplaceable. First, speed and timing are critical when disease threatens crops. An AT-502 treats 250 acres per hour—it would take a full day to treat 250 acres with ground equipment. Second, wet field conditions and rolling terrain often make ground application impossible. Third, aerial application prevents soil compaction and crop damage. Research from Purdue University and Kansas State University shows ground rigs damage 1.5-5% of crops and reduce corn yields by up to 18.6 bushels per acre compared with aerial application.

The agriculture industry's dynamics are shifting dramatically. Globalization has spread crop diseases worldwide faster than ever. Black sigatoka fungus, which devastates banana crops, now requires weekly aerial fungicide applications year-round in tropical regions, a dramatic increase from the handful of treatments needed before widespread outbreak. Farm consolidation is driving demand for larger, more capable aircraft, too. Nearly 40% of Brazilian farms exceed 1,000 acres, compared with less than 10% in the U.S., and Latin America's agricultural aircraft fleet is growing faster than four times the rate of North America's.

Brazil illustrates the market's evolution. While 60% of the country's 2,500+ agricultural fleet are Embraer's piston-powered Ipanemas, operators are upgrading to turbine imports. Embraer delivered 65 Ipanemas in 2023, but the turboprop segment is growing faster as year-round growing seasons and diseases like black sigatoka make turbine aircraft economics compelling. With utilization averaging 500-800 hr. annually, and sometimes exceeding 1,000 hr. in tropical regions, the productivity advantage of turbine aircraft justifies their higher acquisition costs.

Uncrewed aircraft systems (UAS) are the proverbial "elephant in the hangar." More than 2,000 agricultural drones operate in Brazil alone, and drone advocates assert that UAS will replace crewed aircraft. The data suggests otherwise. Even advanced agricultural drones cover just 12-52 acres per hour with 8-80-gal. capacity, requiring constant recharging and refilling. In contrast, an AT-502 carries 500 gal. and treats 250 acres hourly. Simple physics, payload capacity and battery technology mean UAS will remain complementary tools for precision spot-spraying and monitoring rather than replacing crewed aircraft that cover large acreage quickly.

Beyond the Americas, opportunities exist in Africa and the Asia-Pacific region despite infrastructure and regulatory hurdles. Africa's nascent commercial agriculture industry and vast untapped arable land represent long-term potential, and the Philippines' expanding banana plantations face the same black sigatoka pressures driving Latin American demand for aerial application.

Dylan Volanth is a senior associate at AeroDynamic Advisory, with experience across commercial, business and general aviation.

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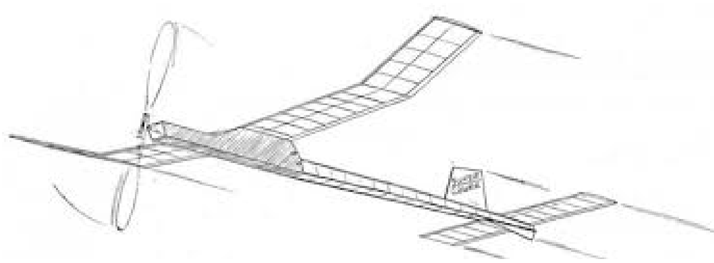

WHAT'S HAPPENING -

July 19, 2026 - San Diego Orbiteer Outdoor Monthly
Taibi Flying Field, Perris CA, 7:30 start time.
<u>Events</u>: P-30 / Glider / Power